

Projectile Motion

Master horizontal and oblique projection for JEE Physics

🕒 6 min read Updated Aug 2026

#kinematics

#projectile

#trajectory

Prerequisites

Before studying this topic, make sure you understand:

- [Motion in a Plane](#) - 2D vector motion
- [Kinematic Equations](#) - for uniform acceleration

What is Projectile Motion?

A **projectile** is any object that is thrown into the air and moves under the influence of gravity alone (no air resistance).

💡 Think Spider-Man!

Remember when **Spider-Man** swings between buildings in *No Way Home* or *Across the Spider-Verse*? When he releases his web at the highest point of his swing and flies through the air — that's projectile motion! His path traces a perfect parabola until he shoots another web.

Key Characteristics:

- Horizontal motion: **uniform velocity** (no acceleration)
- Vertical motion: **uniformly accelerated** ($a = g$ downward)
- Path: **Parabola**

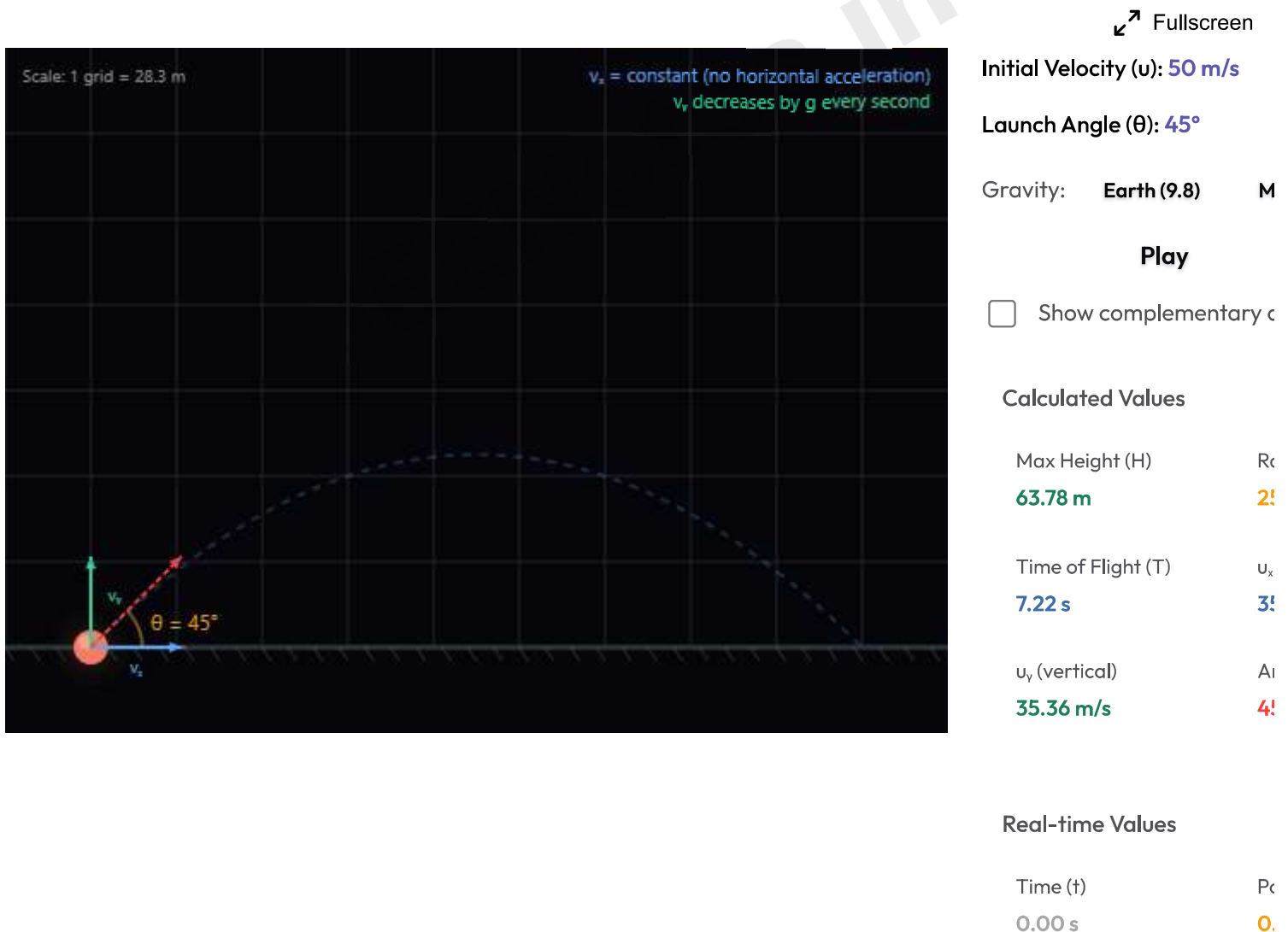
① Independence Principle

The horizontal and vertical motions are **completely independent**. What happens in x doesn't affect y , and vice versa.

Movie Connection: In *Pathaan* (2023), when Shah Rukh Khan jumps from helicopters or between buildings, his horizontal speed stays constant while gravity pulls him down — classic projectile motion!

Interactive Demo

Explore projectile motion with this full-featured simulator. Adjust parameters and observe how they affect the trajectory:



Position y	v_x
0.00 m	3!
v_y (changes)	S_f
35.36 m/s	50
	v_x
	v_y

🔗 How to Use This Animation

Controls:

- **Initial Velocity (u):** Drag the slider to set launch speed (10-100 m/s)
- **Launch Angle:** Set the angle of projection (0-90 degrees)
- **Gravity Toggle:** Switch between Earth ($g = 9.8 \text{ m/s squared}$) and Moon ($g = 1.6 \text{ m/s squared}$)
- **Play/Pause:** Start or pause the animation
- **Reset:** Return projectile to starting position

What to Observe:

Blue arrow (v_x): Horizontal velocity stays **CONSTANT** throughout the flight

Green arrow (v_y): Vertical velocity decreases going up, becomes zero at apex, then increases downward

Red dashed arrow (v): Resultant velocity (vector sum of v_x and v_y)



Green dot at apex: Maximum height where $v_y = 0$

Orange range marker: Horizontal distance covered

Try These Experiments:

1. Set angle to 45 degrees and note the range. Then try 30 degrees and 60 degrees – they have the SAME range!
2. Enable “Show complementary angle” to see both trajectories simultaneously
3. Switch to Moon gravity – notice how much farther the projectile travels
4. At any angle, observe that v_x never changes while v_y constantly decreases

Types of Projectile Motion

1. Horizontal Projection

Object is thrown **horizontally** from a height h with velocity u .

🔗 Pushpa 2 Cliff Scene

In *Pushpa 2: The Rule*, when objects (or people!) are thrown horizontally off a cliff, they don't fall straight down — they travel forward while falling. The horizontal throw determines how FAR they land, while gravity determines how FAST they fall!

Initial Conditions:

- $u_x = u$ (horizontal)
- $u_y = 0$ (no vertical component)

Equations:

Direction	Motion Type	Equations
Horizontal (x)	Uniform	$x = ut$

Direction	Motion Type	Equations
Vertical (y)	Accelerated	$y = \frac{1}{2}gt^2, v_y = gt$

Key Formulas:

$$T = \sqrt{\frac{2h}{g}}$$

$$R = u\sqrt{\frac{2h}{g}} = uT$$

$$v_{final} = \sqrt{u^2 + 2gh}$$

⚡ Time of Flight

Time of flight depends **only on height**, not on horizontal velocity!

2. Oblique Projection (Ground to Ground)

Object is thrown at angle θ with horizontal, with initial velocity u .



Initial Components:

- $u_x = u \cos \theta$
- $u_y = u \sin \theta$

At any time t:

- $x = (u \cos \theta)t$
- $y = (u \sin \theta)t - \frac{1}{2}gt^2$
- $v_x = u \cos \theta$ (constant)
- $v_y = u \sin \theta - gt$

Key Formulas for Oblique Projection

Time of Flight

$$T = \frac{2u \sin \theta}{g}$$

$$\text{Time to reach maximum height} = \frac{T}{2} = \frac{u \sin \theta}{g}$$

Maximum Height

$$H_{max} = \frac{u^2 \sin^2 \theta}{2g}$$

Horizontal Range

$$R = \frac{u^2 \sin 2\theta}{g}$$

Equation of Trajectory

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

This is a **parabola** (quadratic in x).

$$\text{Alternative form: } y = x \tan \theta \left(1 - \frac{x}{R}\right)$$

Important Properties

Maximum Range

Range is maximum when $\sin 2\theta = 1$, i.e., $\theta = 45^\circ$

$$R_{max} = \frac{u^2}{g}$$

🔗 Jawan Train Scene

In *Jawan* (2023), remember when SRK throws objects or jumps between train cars? For maximum horizontal distance, 45° is the magic angle! That's why athletes in long jump or javelin throw aim for approximately this angle.

Complementary Angles

For angles θ and $(90^\circ - \theta)$:

- **Same range** (since $\sin 2\theta = \sin 2(90^\circ - \theta)$)
- Different heights and times

Angle	Range	Height	Time
30°	R	H_1	T_1
60°	R (same)	$H_2 > H_1$	$T_2 > T_1$

Velocity at Any Point

At height h :

$$v = \sqrt{u^2 - 2gh}$$

At any time:

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{u^2 \cos^2 \theta + (u \sin \theta - gt)^2}$$

Velocity Direction

$$\tan \alpha = \frac{v_y}{v_x} = \frac{u \sin \theta - gt}{u \cos \theta}$$

where α is angle with horizontal.

⚠ Common Mistake

Don't confuse:

- Height at time t : $y = u_y t - \frac{1}{2}gt^2$
- Maximum height: $H_{max} = \frac{u^2 \sin^2 \theta}{2g}$

Special Cases

Projection from a Height

If projectile is launched from height h above ground:

Time of flight: Solve $h + (u \sin \theta)T - \frac{1}{2}gT^2 = 0$

$$T = \frac{u \sin \theta + \sqrt{u^2 \sin^2 \theta + 2gh}}{g}$$

Projection on Inclined Plane

For a plane inclined at angle α :

- Range along plane: $R = \frac{2u^2 \sin(\theta - \alpha) \cos \theta}{g \cos^2 \alpha}$
- Maximum range: at $\theta = \frac{\pi}{4} + \frac{\alpha}{2}$

Summary Table

Quantity	Horizontal Projection	Oblique Projection
Initial v_x	u	$u \cos \theta$
Initial v_y	0	$u \sin \theta$
Time of flight	$\sqrt{\frac{2h}{g}}$	$\frac{2u \sin \theta}{g}$
Max height	—	$\frac{u^2 \sin^2 \theta}{2g}$
Range	$u \sqrt{\frac{2h}{g}}$	$\frac{u^2 \sin 2\theta}{g}$

Practice Problems

✔ Problem 1

A ball is thrown horizontally from a 45 m high cliff with speed 20 m/s. Find time of flight and range. ($g = 10 \text{ m/s}^2$)

Solution: $T = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 45}{10}} = \sqrt{9} = 3 \text{ s}$

$R = uT = 20 \times 3 = 60 \text{ m}$

✔ Problem 2

A projectile is launched at 30° with 40 m/s. Find range, max height, and time of flight. ($g = 10 \text{ m/s}^2$)

Solution: $T = \frac{2u \sin \theta}{g} = \frac{2 \times 40 \times 0.5}{10} = 4 \text{ s}$

$H_{max} = \frac{u^2 \sin^2 \theta}{2g} = \frac{1600 \times 0.25}{20} = 20 \text{ m}$

$R = \frac{u^2 \sin 2\theta}{g} = \frac{1600 \times \sin 60^\circ}{10} = \frac{1600 \times 0.866}{10} \approx 138.6 \text{ m}$

✔ Problem 3

For what angle of projection is range equal to maximum height?

Solution: $R = H_{max} \frac{u^2 \sin 2\theta}{g} = \frac{u^2 \sin^2 \theta}{2g} 2 \sin \theta \cos \theta = \frac{\sin^2 \theta}{2} 4 \cos \theta = \sin \theta \tan \theta = 4 \theta = \tan^{-1}(4) \approx 76^\circ$

Related Topics

Within Kinematics

- [Kinematic Equations](#) — The foundation for projectile formulas
- [Motion in a Plane](#) — 2D vectors needed here
- [Relative Velocity](#) — Projectiles observed from moving frames
- [Circular Motion](#) — Another type of 2D motion

Connected Chapters

- [Newton's Laws](#) — Forces during projectile motion
- [Work-Energy](#) — Energy approach to projectile problems
- [Gravitation](#) — Why projectiles follow parabolic paths

Math Connections

- [Trigonometry](#) — Sin, cos for component resolution
- [Parabola](#) — The shape of projectile trajectory
- [Differentiation](#) — For max height, range optimization

Comments

Have a question or spotted a mistake? Join the discussion below.



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